

Result of lab7:-

 D:\DSA Lab works\LAB7(RECURSSION)\Fibonacci.exe

Enter Number to find Fibonacci series: 10

The 10th term of the Fibonacci series is: 55

Process exited after 72.39 seconds with return value 0

Press any key to continue . . .

Result of lab8:-

 D:\DSA Lab works\LAB8(TOWER OF HANOI)\TOH.exe

Enter the number of disks: 3

TOWER OF HANOI

The moves involved are:

Move disk 1 from A to C

Move disk 2 from A to B

Move disk 1 from C to B

Move disk 3 from A to C

Move disk 1 from B to A

Move disk 2 from B to C

Move disk 1 from A to C

Process exited after 14.15 seconds with return value 0

Press any key to continue . . . ■

Result of lab9a:-

```
D:\DSA Lab works\LAB9(BINARY SEARCH TREE)\BST.exe

Binary Search Tree Operations:
1. Insert Node
2. Search Node
3. Delete Node
4. Display BST (Inorder)
5. Inorder Traversal
6. Preorder Traversal
7. Postorder Traversal
0. Exit
Enter your choice: 1
Enter data to insert: 10

Binary Search Tree Operations:
1. Insert Node
2. Search Node
3. Delete Node
4. Display BST (Inorder)
5. Inorder Traversal
6. Preorder Traversal
7. Postorder Traversal
0. Exit
Enter your choice: 1
Enter data to insert: 9
```

```
Binary Search Tree Operations:
1. Insert Node
2. Search Node
3. Delete Node
4. Display BST (Inorder)
5. Inorder Traversal
6. Preorder Traversal
7. Postorder Traversal
0. Exit
Enter your choice: 1
Enter data to insert: 55

Binary Search Tree Operations:
1. Insert Node
2. Search Node
3. Delete Node
4. Display BST (Inorder)
5. Inorder Traversal
6. Preorder Traversal
7. Postorder Traversal
0. Exit
Enter your choice: 1
Enter data to insert: 3
```

Result of lab9b:-

```
Binary Search Tree Operations:
1. Insert Node
2. Search Node
3. Delete Node
4. Display BST (Inorder)
5. Inorder Traversal
6. Preorder Traversal
7. Postorder Traversal
0. Exit
Enter your choice: 1
Enter data to insert: 55
Duplicate key (55) not inserted.
```

```
Binary Search Tree Operations:
1. Insert Node
2. Search Node
3. Delete Node
4. Display BST (Inorder)
5. Inorder Traversal
6. Preorder Traversal
7. Postorder Traversal
0. Exit
Enter your choice: 2
Enter data to search: 1
Node with data 1 not found.
```

```
Binary Search Tree Operations:
1. Insert Node
2. Search Node
3. Delete Node
4. Display BST (Inorder)
5. Inorder Traversal
6. Preorder Traversal
7. Postorder Traversal
0. Exit
Enter your choice: 3
Enter data to delete: 9
```

```
Binary Search Tree Operations:
1. Insert Node
2. Search Node
3. Delete Node
4. Display BST (Inorder)
5. Inorder Traversal
6. Preorder Traversal
7. Postorder Traversal
0. Exit
Enter your choice: 4
Inorder Traversal (Sorted Order): 3 10 55
```

Result of lab9c:-

```
Binary Search Tree Operations:
```

1. Insert Node
2. Search Node
3. Delete Node
4. Display BST (Inorder)
5. Inorder Traversal
6. Preorder Traversal
7. Postorder Traversal
0. Exit

```
Enter your choice: 5
```

```
Inorder Traversal: 3 10 55
```

```
Binary Search Tree Operations:
```

1. Insert Node
2. Search Node
3. Delete Node
4. Display BST (Inorder)
5. Inorder Traversal
6. Preorder Traversal
7. Postorder Traversal
0. Exit

```
Enter your choice: 6
```

```
Preorder Traversal: 10 3 55
```

```
Binary Search Tree Operations:
```

1. Insert Node
2. Search Node
3. Delete Node
4. Display BST (Inorder)
5. Inorder Traversal
6. Preorder Traversal
7. Postorder Traversal
0. Exit

```
Enter your choice: 7
```

```
Postorder Traversal: 3 55 10
```

```
Binary Search Tree Operations:
```

1. Insert Node
2. Search Node
3. Delete Node
4. Display BST (Inorder)
5. Inorder Traversal
6. Preorder Traversal
7. Postorder Traversal
0. Exit

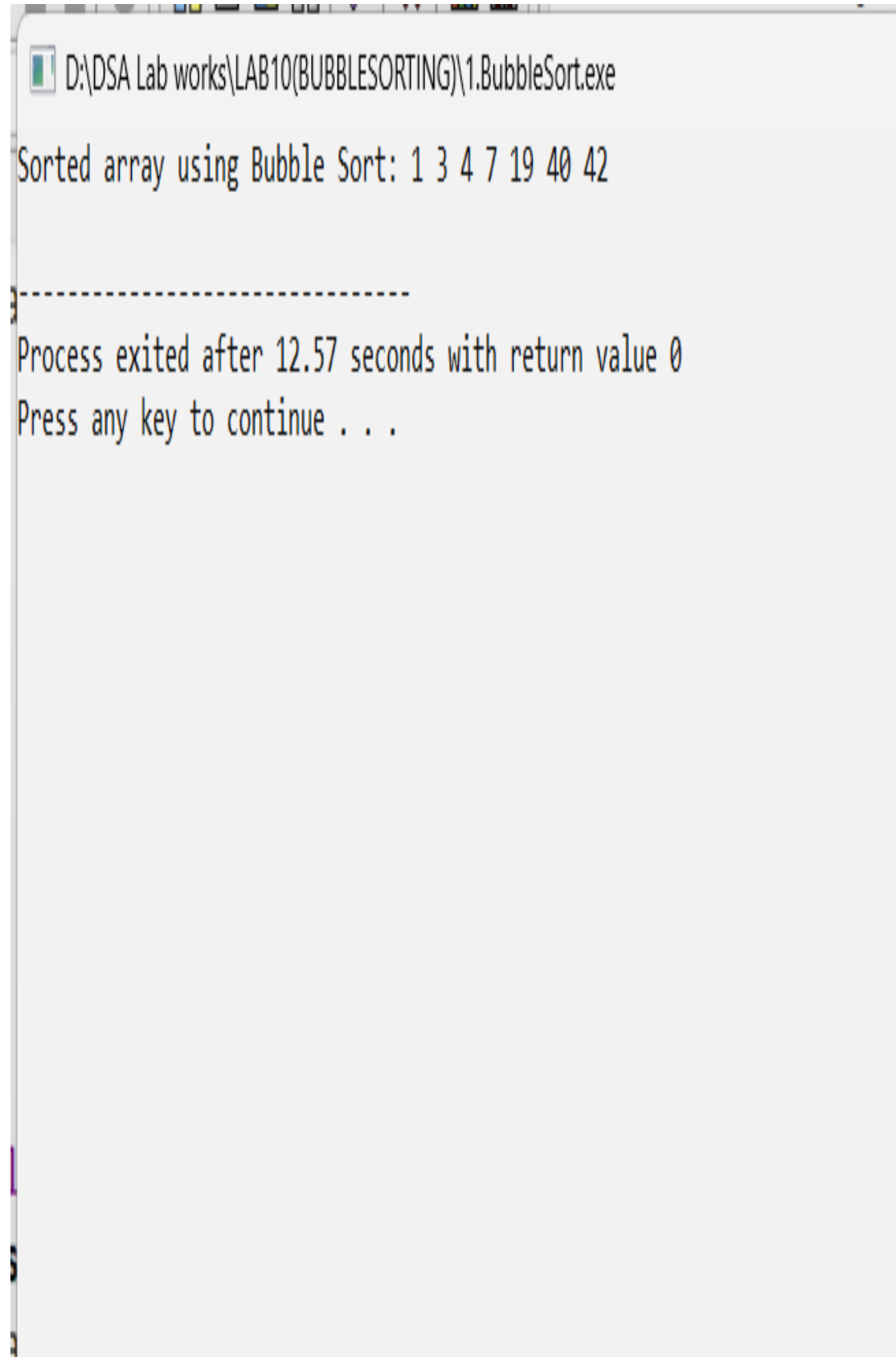
```
Enter your choice: 0
```

```
Exiting program.
```

```
-----  
Process exited after 6805 seconds with return value 0
```

```
Press any key to continue . . .
```

Result of lab10:-



```
D:\DSA Lab works\LAB10(BUBBLESORTING)\1.BubbleSort.exe
Sorted array using Bubble Sort: 1 3 4 7 19 40 42
-----
Process exited after 12.57 seconds with return value 0
Press any key to continue . . .
```

Result of lab11:-

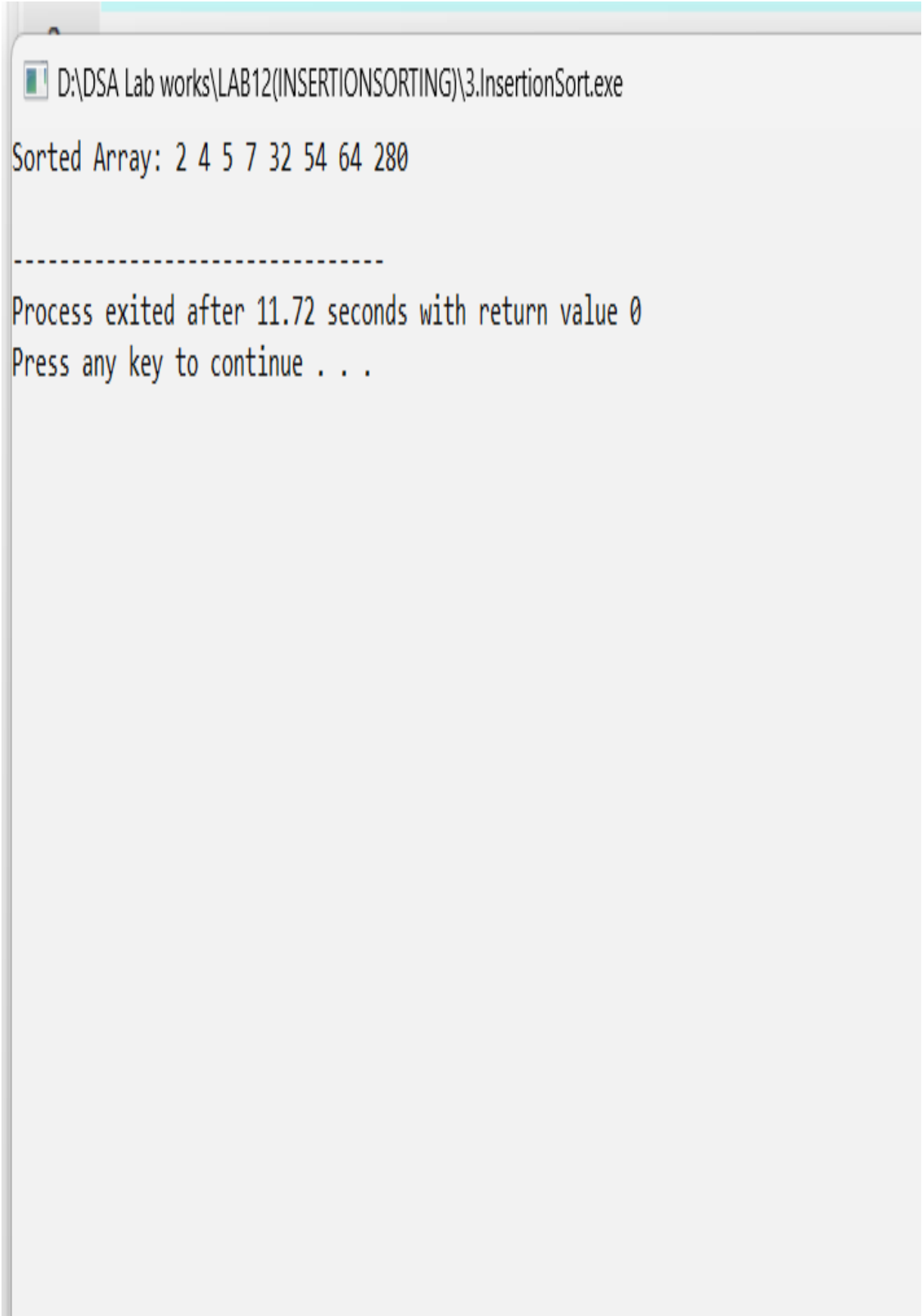
 D:\DSA Lab works\LAB11(SELECTIONSORTING)\2.selectionSort.exe

Sorted Elements: 2 15 20 24 34 53 75

Process exited after 12.41 seconds with return value 0

Press any key to continue . . .

Result of lab12:-



```
D:\DSA Lab works\LAB12(INSERTIONSORTING)\3.InsertionSort.exe
Sorted Array: 2 4 5 7 32 54 64 280
-----
Process exited after 11.72 seconds with return value 0
Press any key to continue . . .
```

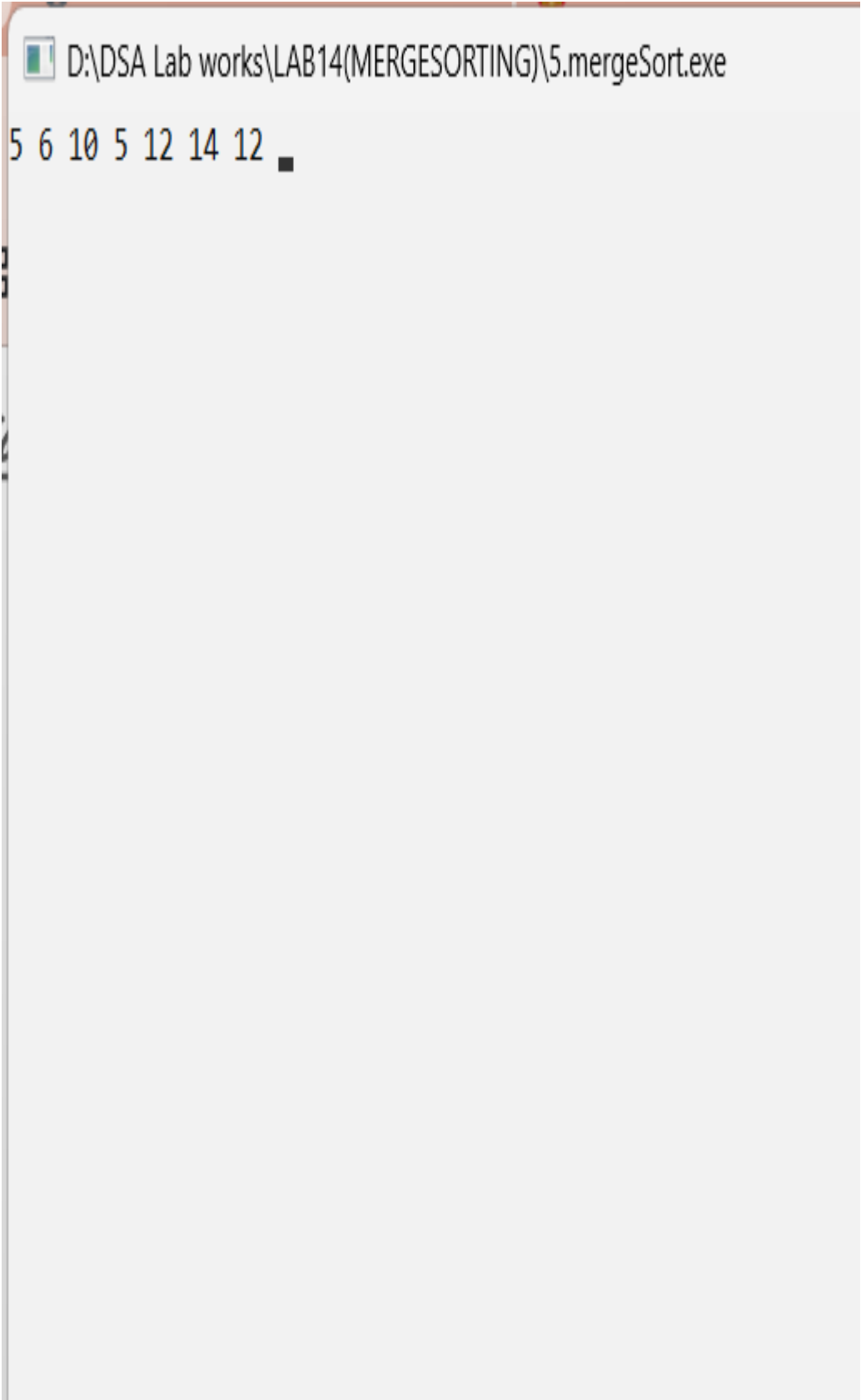

Result of lab13:-



```
D:\DSA Lab works\LAB13(QUICKSORTING)\4.quicksort.exe
Sorted elements:
1 2 3 4 6 8 9
```

The image shows a screenshot of a Windows command prompt window. The title bar at the top reads "D:\DSA Lab works\LAB13(QUICKSORTING)\4.quicksort.exe". The command prompt displays the text "Sorted elements:" followed by a new line and the numbers "1 2 3 4 6 8 9" on the next line. The window has a standard Windows XP-style title bar with minimize, maximize, and close buttons.

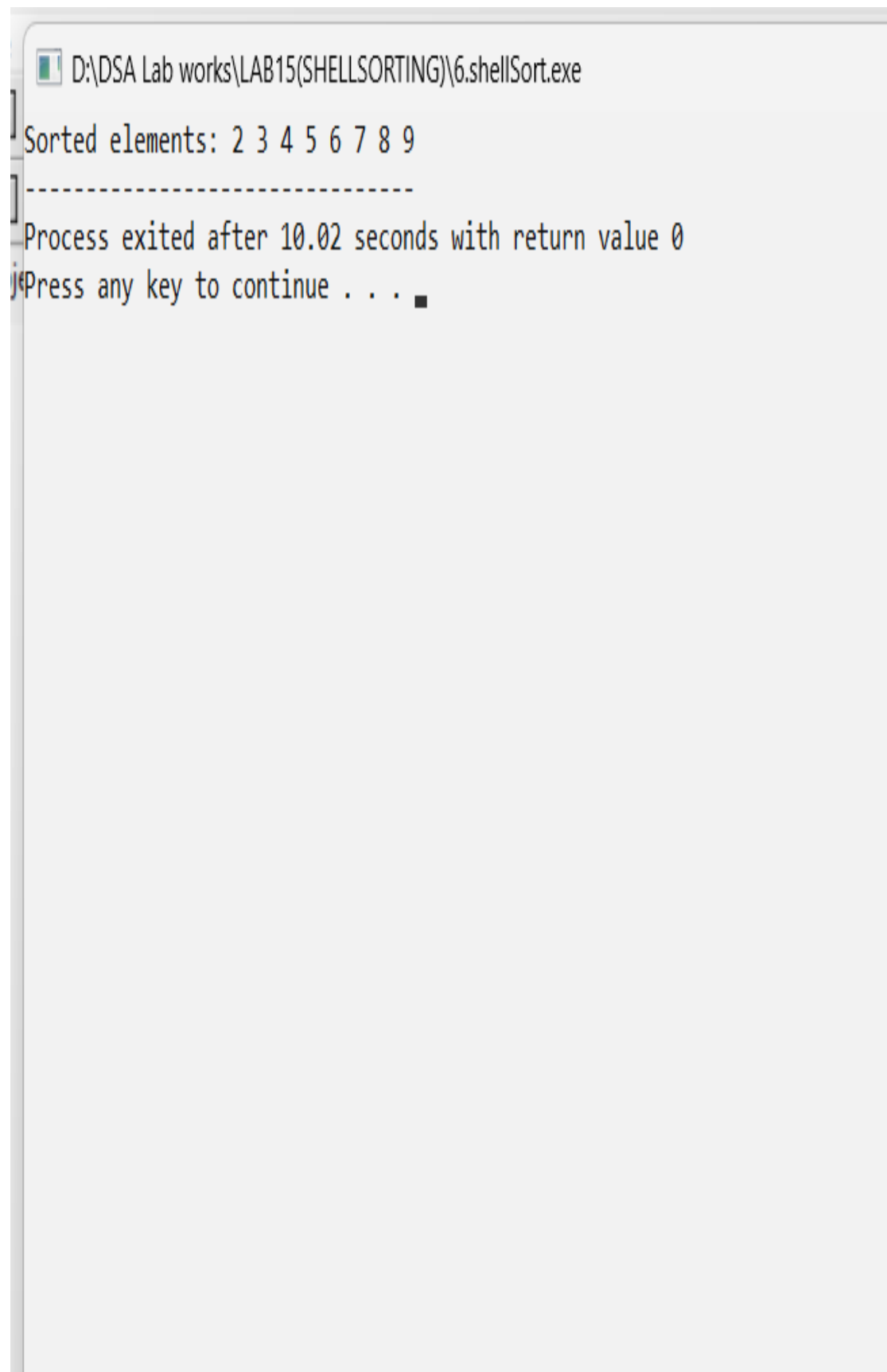
Result of lab14:-



The screenshot shows a Windows command prompt window with the title bar "D:\DSA Lab works\LAB14(MERGESORTING)\5.mergeSort.exe". The command prompt displays the input sequence "5 6 10 5 12 14 12" followed by a cursor. The window has a standard Windows interface with a taskbar at the bottom.

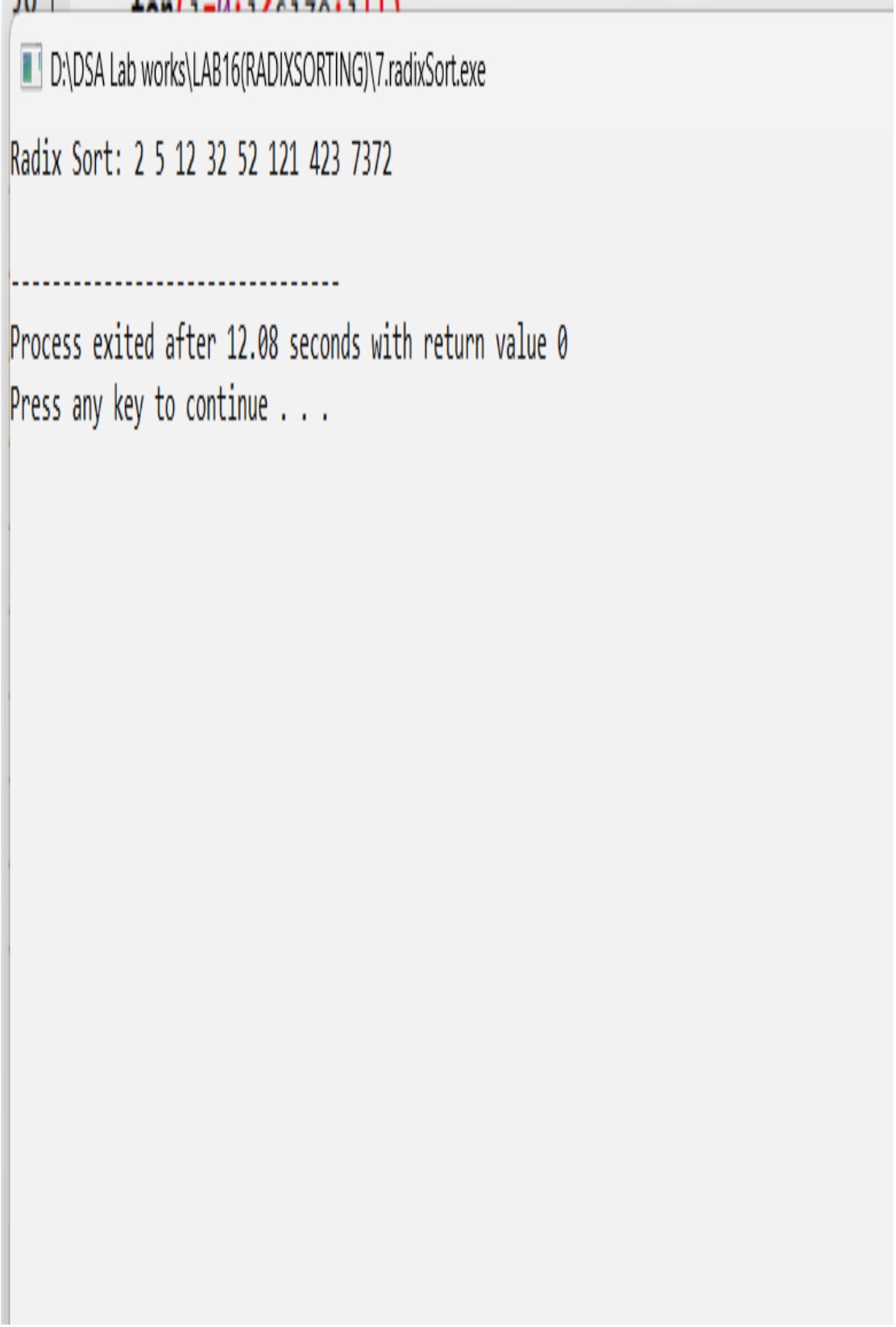
```
D:\DSA Lab works\LAB14(MERGESORTING)\5.mergeSort.exe  
5 6 10 5 12 14 12 █
```

Result of lab15:-



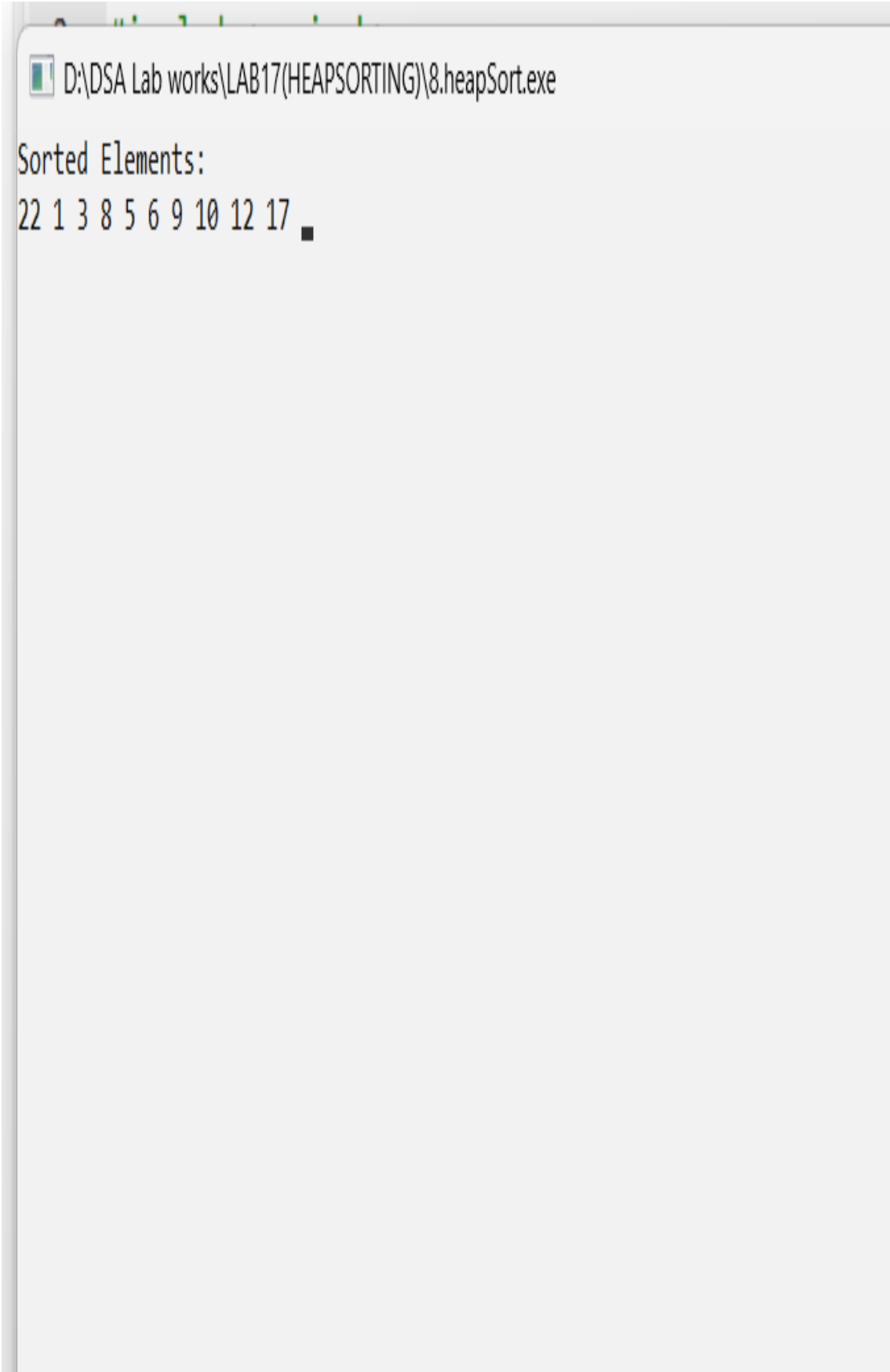
```
D:\DSA Lab works\LAB15(SHELLSORTING)\6.shellSort.exe
Sorted elements: 2 3 4 5 6 7 8 9
-----
Process exited after 10.02 seconds with return value 0
Press any key to continue . . .
```

Result of lab16:-



```
D:\DSA Lab works\LAB16(RADIXSORTING)\7.radixSort.exe
Radix Sort: 2 5 12 32 52 121 423 7372
-----
Process exited after 12.08 seconds with return value 0
Press any key to continue . . .
```

Result of lab17:-



The screenshot shows a Windows command prompt window with the title bar "D:\DSA Lab works\LAB17(HEAPSORTING)\8.heapSort.exe". The command prompt displays the text "Sorted Elements:" followed by the numbers "22 1 3 8 5 6 9 10 12 17" and a cursor. The numbers are color-coded: "22" is blue, "1" is red, "3" is green, "8" is blue, "5" is red, "6" is green, "9" is blue, "10" is red, "12" is green, and "17" is blue. A small black square cursor is positioned at the end of the output.

```
D:\DSA Lab works\LAB17(HEAPSORTING)\8.heapSort.exe  
Sorted Elements:  
22 1 3 8 5 6 9 10 12 17 ■
```

Result of lab18:-

```
D:\DSA Lab works\LAB18(SEARCHING-HASHING)\Hashing.exe
10 inserted at arr[3]
4 inserted at arr[4]
2 inserted at arr[2]
Collision occured !!! arr[3] has element 10 already!
Collision occured !!! arr[3] has element 10 already!
Hash table
arr[0]=-1
arr[1]=-1
arr[2]=2
arr[3]=10
arr[4]=4
arr[5]=3
arr[6]=17

Deleting value 10...
After deletion hash table
arr[0]=-1
arr[1]=-1
arr[2]=2
arr[3]=-1
arr[4]=4
arr[5]=3
arr[6]=17

Deleting value 5...
5 not present in the hash table
After deletion hash table
arr[0]=-1
arr[1]=-1
arr[2]=2
arr[3]=-1
arr[4]=4
arr[5]=3
arr[6]=17

Searching value 4...
Search found.
Searching value 10...
Search not found.

-----
Process exited after 11.83 seconds with return value 0
Press any key to continue . . .
```